

Spring 2026. Physics 510.

Quantum Theory II.

Homework # 1.

Due to February 4th, 2026.

1. **Shift of creation and annihilation operators.** Determine if the transformation from operators of creation and annihilation $a^\dagger$ and a to the new operators $b^\dagger$ and b , where $b = a + \alpha$ and α is a complex number, is the unitary one. Discuss both fermionic and bosonic case.

2. **More Bogolyubov transformations.** Consider the Hamiltonian operator $\mathcal{H} = \epsilon(a_1^\dagger a_1 + a_2^\dagger a_2) + \lambda(a_1^\dagger a_2^\dagger + a_2 a_1)$ where $a_1^\dagger, a_1$ etc. are boson operators of creation and annihilation. Use the Bogolyubov transformation in the following form

$$\begin{aligned}a_1^\dagger &= ub_1^\dagger + vb_2 \\ a_2^\dagger &= ub_2^\dagger + vb_1\end{aligned}$$

Calculate parameters u and v to diagonalize the Hamiltonian and find out the energy of new particles and the vacuum energy. Hint: recall what we did in the class for fermions.

3. **Nonlinear "Bogolyubov transformations".** Suppose Fermi annihilation and creation operators a and $a^\dagger$ satisfy the standard anticommutation relations. Find all the transformations that preserve those relations. Hint: the transformations must not be linear! Explain why the transformation $b = ua + va^\dagger + wa^\dagger a + q$ is the most general form of such transformations and then find its coefficients.

4. **Toy model of superconductivity.**

a) Consider a system of fermions with two single-particle states $|\mathbf{k}\rangle$ and $|\mathbf{-k}\rangle$ corresponding to the creation operators, $a_{\mathbf{k}}^\dagger$ and $a_{\mathbf{-k}}^\dagger$.

b) Introduce new fermion operators defined as following:

$$b_{\mathbf{k}} = u_{\mathbf{k}} a_{\mathbf{k}} + v_{\mathbf{k}} a_{\mathbf{-k}}^\dagger; \quad b_{\mathbf{-k}} = u_{\mathbf{-k}} a_{\mathbf{-k}} + v_{\mathbf{-k}} a_{\mathbf{k}}^\dagger,$$

where $u_{\mathbf{k}}, v_{\mathbf{k}}, u_{\mathbf{-k}},$ and $v_{\mathbf{-k}}$ are real coefficients.

c) Using anticommutators of new fermion operators $b_{\mathbf{k}}, b_{\mathbf{-k}}, b_{\mathbf{k}}^\dagger,$ and $b_{\mathbf{-k}}^\dagger$ derive the equations for the coefficients.

d) Find the coefficients out. **Hint:** they should depend on a single arbitrary parameter $\vartheta_{\mathbf{k}}$. Equations for pairs $u_{\mathbf{k}}, v_{\mathbf{k}}$ and $u_{\mathbf{-k}}, v_{\mathbf{-k}}$ are the same as we spoke about at the class earlier. However, an additional equation that mixes those coefficients will let you determine the trigonometric functions correctly.

e) Calculate how operators $a_{\mathbf{k}}^\dagger$ and $a_{\mathbf{k}}$ depend on operators $b_{\mathbf{k}}, b_{\mathbf{-k}}, b_{\mathbf{k}}^\dagger,$ and $b_{\mathbf{-k}}^\dagger$.

f) Introduce a NEW vacuum $|0\rangle$ defined as $b_{\mathbf{k}}|0\rangle,$ and $b_{\mathbf{-k}}|0\rangle$.

g) Evaluate the vacuum expectation value of $\langle 0 | a_{\mathbf{k}}^\dagger a_{\mathbf{k}} | 0 \rangle$.

5. **Supersymmetric model. The simplest one. Abers book. Problem 13.3.** See the separate file. For those who are interested in the supersymmetry theory: you can try problem 13.4.